

MULTICOLLINEARITY

1) What is it?

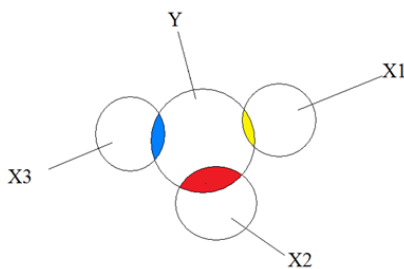
When two or more covariates in a model are highly correlated.

When one covariate is linearly expressed by other covariates in the model (a linear function)

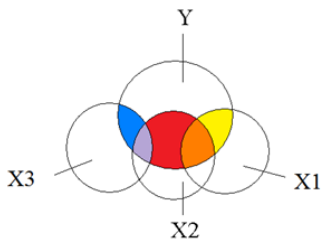
$$Y = X1 + X2 + X3$$

Red, blue and yellow are where each X explains variance in Y

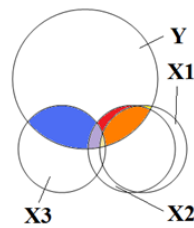
No multicollinearity:



Moderate:



Extreme:



[What is Multicollinearity? A Visual Description \(theanalysisfactor.com\)](https://theanalysisfactor.com/what-is-multicollinearity-a-visual-description/)

- 2) Is it a problem? May not affect the accuracy of the model but might make interpreting the effects for individual predictors problematic.
- 3) Can lead to skewed or misleading results when interpreting independent variable(s)
 - a. Large standard errors -- wide confidence intervals -- less reliable probabilities
 - b. Imprecise or nonsensical estimates: 0 or huge
- 4) Suspect when:
 - a. Large standard errors
 - b. Regression coefficients change when predictor variables are added or deleted
 - c. Regression coefficients not significant in a multivariable regression model but significant in a simple linear regression (unexpected results)
 - d. Coefficients opposite of what was expected (unexpected results)
 - e. R^2 for the model is high but no/few significant predictors

- 5) Detect using:
 - a. Variance inflation factor (VIF) - includes all variables in the model
 - b. Correlation –bivariate relationship between two variables
 - c. Condition indices²
- 6) Reduced by:
 - a. Remove variable(s) or combine or transform (mean centering, standardize)
 - b. Ridge regression or partial squares regression or principal component regression

VIF

- 1) Represents how well the variable is explained by the other independent variables.¹
- 2) Measures the strength of the correlation between predictor variables. Regression model is fit using each independent variable (IV) as the dependent and all other variables as predictors

$X_1 = X_2, X_3, X_4 \rightarrow R^2$
 $X_2 = X_1, X_3, X_4 \rightarrow R^2$
 $X_3 = X_1, X_2, X_4 \rightarrow R^2$
 $X_4 = X_1, X_2, X_3 \rightarrow R^2$

R^2 indicates the percentage of variance in the individual IV that the set of IVs explains

$$VIF_i = \frac{1}{1 - R_i^2}$$

General rule of thumb

1 = not correlated

2-4 = moderately correlated. May/not require attention.

5+ = highly correlated*

* some use 2.5 others 10

A VIF of 4.0 means that the standard error for the coefficient of that predictor variable is 4.0 times larger than if that predictor variable had 0 correlation with the other predictor variables.
(Source: Statology)

STATA EXAMPLE: LINEAR REGRESSION

global labs VARLIST

global labr REVISED_VARLIST

asdoc regress adl_difficulty_num \$labs, title(Table 1: Regression results)

save(H:\Stats\multicollinearity\results tables.rtf) replace

asdoc vif, label title(Table 2: Multicollinearity) save(H:\Stats\multicollinearity\results tables.rtf) append

asdoc pwcorr VARIABLES, label save(H:\Stats\multicollinearity\results tables.rtf) title(Table 3: Pearson Correlations) append

asdoc regress adl_difficulty_num \$labr, title(Table 4: Revised regression results with variables removed)

save(H:\Stats\multicollinearity\results tables.rtf) append

asdoc vif, label title(Table 5: Multicollinearity revised) save(H:\Stats\multicollinearity\results tables.rtf) append

Table 1: Regression results

adl_difficulty_num	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]
msas_phys_num_pres~b	-1.245	1.659	-0.75	.507	-6.524	4.034
mortality_score	-.018	.131	-0.14	.899	-.436	.400
eligibility_bun_10	-.023	.064	-0.36	.741	-.226	.180
eligibility_sbp_10	-.050	.129	-0.39	.725	-.459	.360
eligibility_cr_10	.108	.332	0.33	.766	-.947	1.1630
mean_life_expectancy	.238	.379	0.63	.574	-.966	1.443
heart_failure_adm~10	.655	.700	0.94	.418	-1.573	2.883
age_10	-.212	1.293	-0.16	.880	-4.326	3.903
bun_units1_11	.089	.106	0.84	.462	-.248	.426
pain_bother_b	.925	1.246	0.74	.512	-3.041	4.892
nausea_bother_b	.270	.983	0.27	.802	-2.86	3.400
kccq_domain_100	.097	.106	0.92	.425	-.239	.433

Table 2: Multicollinearity

	VIF	1/VIF
bun units1 11	9.228	.108
MSAS num physical symptoms	9.225	.108
kccq domain 100	9.072	.11
SBP at time of study eligibility	7.267	.138
Seattle heart failure score	5.702	.175
Number of heart failure admits	5.453	.183
Mean life expectancy	4.472	.224
Pain bother b	3.047	.328
age 10	2.989	.335
Creatinine	2.259	.443
BUN	2.135	.468
Nausea bother b	2.012	.497
Mean VIF	5.239	.

Table 3: Pearson Correlations

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) msas_phys_num_~b	1.000										
(2) mortality_score	-0.064	1.000									
(3) eligibility_b~10	0.011	0.157*	1.000								
(4) eligibility_s~10	0.064	-0.030	0.046	1.000							
(5) eligibility_c~10	0.016	0.008	0.277*	0.045	1.000						
(6) mean_life_expe~y	-0.019	-0.405*	-0.349*	0.206*	-0.014	1.000					
(7) heart_failure~10	0.168*	0.072	-0.054	-0.016	-0.103*	-0.064	1.000				
(8) age_10	-0.157*	0.137*	0.224*	0.059	0.034	-0.257*	-0.221*	1.000			
(9) bun_units1_11	0.121	0.189*	0.756*	0.201*	0.177*	-0.366*	-0.030	0.204*	1.000		
(10) pain_bother_b	0.260*	-0.071	-0.062	0.050	-0.065	0.050	0.234*	-0.130	-0.019	1.000	
(11) nausea_bother_b	0.214	0.344*	0.180	-0.048	-0.009	-0.072	-0.045	-0.120	-0.051	0.008	1.000
(12) kccq_domain_100	0.534*	-0.002	-0.070	0.038	0.034	0.029	0.152*	-0.211*	0.002	0.202*	0.308*

Table 4 Revised: Regression results - bun_units1_11 removed from model

adl_difficulty_num	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]
msas_phys_num_pres~b	-.038	.284	-0.13	.894	-.621	.545
mortality_score	-.025	.023	-1.11	.275	-.072	.021
eligibility_bun_10	-.002	.023	-0.11	.916	-.051	.046
eligibility_sbp_10	.006	.025	0.25	.803	-.044	.057
eligibility_cr_10	.139	.145	0.96	.347	-.158	.436
mean_life_expectancy	-.025	.116	-0.22	.831	-.263	.213
heart_failure_adm~10	.237	.179	1.32	.197	-.103	.604
age_10	.021	.362	0.06	.954	-.722	.764
pain_bother_b	.393	.368	1.07	.296	-.363	1.148
nausea_bother_b	.350	.394	0.89	.382	-.459	1.159
kccq_domain_100	.010	.018	0.54	.594	-.027	.046

Table 1: Regression results – FOR COMPARISON

adl_difficulty_num	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]
msas_phys_num_pres~b	-1.245	1.659	-0.75	.507	-6.524	4.034
mortality_score	-.018	.131	-0.14	.899	-.436	.400
eligibility_bun_10	-.023	.064	-0.36	.741	-.226	.180
eligibility_sbp_10	-.050	.129	-0.39	.725	-.459	.360
eligibility_cr_10	.108	.332	0.33	.766	-.947	1.1630
mean_life_expectancy	.238	.379	0.63	.574	-.966	1.443
heart_failure_adm~10	.655	.700	0.94	.418	-1.573	2.883
age_10	-.212	1.293	-0.16	.880	-4.326	3.903
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pain_bother_b	.925	1.246	0.74	.512	-3.041	4.892
nausea_bother_b	.270	.983	0.27	.802	-2.86	3.400
kccq_domain_100	.097	.106	0.92	.425	-.239	.433

Table 5 Revised : Multicollinearity - variable removed from model

	VIF	1/VIF
Nausea bother b	2.082	.480
MSAS num physical symptoms prese	1.932	.518
age 10	1.739	.575
Seattle heart failure score (fro	1.621	.617
Mean life expectancy ate eligibi	1.577	.634
BUN at time of stidy eligibility	1.576	.634
kccq domain 100	1.550	.645
Pain bother b	1.458	.686
Creatinine at time of study elig	1.340	.746
Number of heart failure admissio	1.335	.749
SBP at time of study eligibility	1.219	.821
Mean VIF	1.584	.

STATA OPTIONS FOR DICHOTOMOUS & CATEGORICAL COVARIATES:

- 1) Chi-square test or Kappa agreement
- 2) Tetrachoric or polychoric correlations

```
asdoc tabulate lack_of_energy_present_b lack_of_appetite_present_b, chi nokey column
save(H:\Stats\multicollinearity\results tables.rtf) title(Table 5: Energy & Appetite) append
asdoc, text(Pearson chi2 = 58.5677 Pr = 0.000) save(H:\Stats\multicollinearity\results tables.rtf)
append
```

```
asdoc kap lack_of_energy_present_b lack_of_appetite_present_b,
save(H:\Stats\multicollinearity\results tables.rtf) append
```

```
tetrachoric DICHOTOMOUS_VARIABLES, star(0.05) pw
matrix tet = r(Rho)
putexcel set " H:\Stats\multicollinearity\results tables.rtf \correl table.xlsx", sheet(Tetrachoric) replace
putexcel A3 = matrix(tet), names nformat(number_d2)
```

```
polychoric CATEGORICAL_VARIABLES
matrix poly = r(R)
putexcel set " H:\Stats\multicollinearity\results tables.rtf \correl table.xlsx ", sheet(Polychoric) modify
putexcel A3 = matrix(poly), names nformat(number_d2)
```

Table 6: Energy & Appetite

Lack_of_energy_present_b	Lack_of_appetite_present_b		
	0	1	Total
0	150	10	160
	50.00	8.85	38.74
1	150	103	253
	50.00	91.15	61.26
Total	300	113	413
	100.00	100.00	100.00

First row has *frequencies* and second row has *column percentages*

Pearson chi2 = 58.5677 Pr = 0.000

Expected

Agreement agreement Kappa Std. err. Z Prob>Z
 61.26% 44.90% 0.2969 0.0388 7.65 0.0000

Table 7: Energy & Concentration

Lack_of_energy_present_b	Diff_concentrating_present_b		
	0	1	Total
0	150	10	160
	48.39	9.71	38.74
1	160	93	253
	51.61	90.29	61.26
Total	310	103	413
	100.00	100.00	100.00

First row has *frequencies* and second row has *column percentages*

Pearson chi2 = 48.7354 Pr = 0.000

Expected

Agreement agreement Kappa Std. err. Z Prob>Z
 58.84% 44.36% 0.2602 0.0373 6.98 0.0000

Table 8: Tetrachoric Correlations between Dichotomous Variables

	pain	energy	sob	appetite	nausea	drowsy	Dry mouth	constip	Diff sleep	Diff concent	Weight loss
pain	1.00										
Lack of energy	0.40	1.00									
Sob	0.30	0.57	1.00								
Lack of appetite	0.41	0.66	0.46	1.00							
nausea	0.32	0.45	0.38	0.49	1.00						
drowsy	0.33	0.57	0.46	0.43	0.52	1.00					
Dry mouth	0.26	0.34	0.31	0.35	0.25	0.31	1.00				
constipation	0.10	0.14	0.05	0.26	0.34	0.28	0.28	1.00			
Diff sleeping	0.32	0.44	0.31	0.47	0.52	0.39	0.35	0.18	1.00		
Diff concentrating	0.33	0.63	0.34	0.50	0.49	0.46	0.44	0.25	0.48	1.00	
Weight loss	0.19	0.20	0.15	0.45	0.33	0.23	0.13	0.17	0.08	0.21	1.00

STATA OTHER OPTIONS

Collin VARLIST – compute VIF, condition index, Eigen values

coldiag2 VARLIST – compute condition index

References:

¹[What is Multicollinearity? Here's Everything You Need to Know | by Aniruddha Bhandari | Analytics Vidhya | Medium](#)

²[HG Notes-Identification of Multicollinearity-VIF and Conditioning Number 20140304 \(insead.edu\)](#)

[Eight Ways to Detect Multicollinearity - The Analysis Factor](#)

[What is Multicollinearity? A Visual Description \(theanalysisfactor.com\)](#)